

Commerce XI Maths App

Miscellaneous Exercise 2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 2 - Functions | Total questions: 17

Question 1 - Q1(i) Function

Consider $R = \{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$. Is R a function?

Options

- A. Yes
- B. No, because two inputs have one image
- C. No, because 14 is too large
- D. No, because the range has seven elements

Correct answer: Yes

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives Yes. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 2 - Q1(i) Domain

For $R = \{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$, determine the domain.

Options

- A. $\{2, 4, 6, 8, 10, 12, 14\}$
- B. $\{1, 2, 3, 4, 5, 6, 7\}$
- C. $\{2, 4, 6, 8, 10, 12\}$
- D. $\{1, 2, 3, 4, 5, 6, 7, 14\}$

Correct answer: $\{2, 4, 6, 8, 10, 12, 14\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $\{2, 4, 6, 8, 10, 12, 14\}$. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 3 - Q1(i) Range

For $R = \{(2, 1), (4, 2), (6, 3), (8, 4), (10, 5), (12, 6), (14, 7)\}$, determine the range.

Options

- A. $\{1, 2, 3, 4, 5, 6, 7\}$
- B. $\{2, 4, 6, 8, 10, 12, 14\}$
- C. $\{1, 2, 3, 4, 5, 6\}$
- D. $\{2, 3, 4, 5, 6, 7\}$

Correct answer: $\{1, 2, 3, 4, 5, 6, 7\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives {1, 2, 3, 4, 5, 6, 7}. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 4 - Q1(ii)

Is $\{(0, 0), (1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3), (16, 4), (16, -4)\}$ a function?

Options

- A. No, because some first components have two images
- B. Yes, it is a function
- C. No, because 0 is used
- D. No, because the second components include negatives

Correct answer: No, because some first components have two images

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives No, because some first components have two images. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 5 - Q1(iii) Function

Is $\{(1, 1), (3, 1), (5, 2)\}$ a function?

Options

- A. Yes
- B. No, because 1 is the image of two inputs
- C. No, because 2 has no pre-image
- D. No, because the domain has three elements

Correct answer: Yes

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives Yes. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 6 - Q1(iii) Domain

For $f = \{(1, 1), (3, 1), (5, 2)\}$, determine the domain.

Options

- A. {1, 3, 5}
- B. {1, 2}
- C. {1, 2, 3, 5}

D. $\{1, 1, 2\}$

Correct answer: $\{1, 3, 5\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $\{1, 3, 5\}$. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 7 - Q1(iii) Range

For $f = \{(1, 1), (3, 1), (5, 2)\}$, determine the range.

Options

A. $\{1, 2\}$

B. $\{1, 3, 5\}$

C. $\{1, 1, 2\}$

D. $\{1, 2, 3, 5\}$

Correct answer: $\{1, 2\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $\{1, 2\}$. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 8 - Q2

For $f(x) = 3x/5 + 2$, $x \in \mathbb{R}$, find $f^{-1}(x)$.

Options

A. $(5x - 10)/3$

B. $(3x - 2)/5$

C. $5x/3 + 2$

D. $(5x + 10)/3$

Correct answer: $(5x - 10)/3$

Explanation and steps

Let $y = 3x/5 + 2$. Then $y - 2 = 3x/5$, so $x = 5(y - 2)/3$. Replace y by x to obtain $f^{-1}(x) = (5x - 10)/3$.

Question 9 - Q3 $f(-1)$

A function is defined by $f(x) = 4x + 5$ for $-4 \leq x < 0$. find $f(-1)$.

Correct answer: 1

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives 1. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 10 - Q3 $f(-2)$

A function is defined by $f(x) = 4x + 5$ for $-4 \leq x < 0$. find $f(-2)$.

Correct answer: -3

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives -3. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 11 - Q3 $f(0)$

A function is defined by $f(x) = 4x + 5$ for $-4 \leq x < 0$. What is $f(0)$?

Options

- A. It does not exist
- B. 5
- C. 0
- D. -4

Correct answer: It does not exist

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives It does not exist. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 12 - Q4

A function is defined by $f(x) = 5 - x$ for $0 \leq x \leq 4$. Find x if $f(x) = 3$.

Correct answer: 2

Explanation and steps

$5 - x = 3$ gives $x = 2$, which lies in the given domain $0 \leq x \leq 4$.

Question 13 - Q5

If $f(x) = 3x^2 - 5x + 7$, find $f(x - 1)$.

Options

- A. $3x^2 - 11x + 15$
- B. $3x^2 - 5x + 3$
- C. $3x^2 - 8x + 7$

D. $3x^2 - 11x + 9$

Correct answer: $3x^2 - 11x + 15$

Explanation and steps

Replace x by $x - 1$: $3(x - 1)^2 - 5(x - 1) + 7 = 3x^2 - 11x + 15$.

Question 14 - Q6

If $f(x) = 3x + a$ and $f(1) = 7$, find a and $f(4)$.

Options

A. $a = 4, f(4) = 16$

B. $a = 4, f(4) = 12$

C. $a = 3, f(4) = 16$

D. $a = 10, f(4) = 7$

Correct answer: $a = 4, f(4) = 16$

Explanation and steps

From $f(1) = 3 + a = 7$, $a = 4$. Then $f(4) = 12 + 4 = 16$.

Question 15 - Q7

If $f(x) = ax^2 + bx + 2$, $f(1) = 3$ and $f(4) = 42$, find a and b .

Options

A. $a = 3, b = -2$

B. $a = -2, b = 3$

C. $a = 2, b = -1$

D. $a = 3, b = 2$

Correct answer: $a = 3, b = -2$

Explanation and steps

The conditions give $a + b = 1$ and $16a + 4b = 40$. Dividing the second by 4 gives $4a + b = 10$. Hence $3a = 9$, so $a = 3$ and $b = -2$.

Question 16 - Q8

If $f(x) = (2x - 1)/(5x - 2)$, $x \neq 2/5$, then $(f \circ f)(x)$ equals

Options

A. x

B. $1/x$

C. $(2x + 1)/(5x + 2)$

D. 0

Correct answer: x

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives x. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 17 - Q9

If $f(x) = (x + 3)/(4x - 5)$ and $g(x) = (3 + 5x)/(4x - 1)$, then $(f \circ g)(x)$ equals

Options

A. x

B. $1/x$

C. $x + 3$

D. $4x - 1$

Correct answer: x

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives x. The other choices use an incorrect substitution, operation, domain restriction, or function order.